Nuclear story

In 1989, I led the EPRI-NSF conference on cold fusion, for which I now find a copy of the proceedings online.

https://www.lenr-canr.org/acrobat/EPRInsfepriwor.pdf

Julian Schwinger and Ed Teller both came to see me (and in Teller's case my boss) to urge us not to make public what we learned.

In Teller's case, we promised to cooperate with government attempts to prevent the full information from leaking out..

until and unless circumstances changed, in which case we should use our own best judgment. Circumstances HAVE changed,

and demand that more key people know the story and the larger implications.

One of the changes is that Larry Page has committed many millions of dollars to a new "clean energy" project

which basically takes over a project from Mitsubishi which I have tracked a lot. I have downloaded many files which

I found quite alarming, giving technical details. The clock is now ticking. Another change is that we have a growing

need to be able to "see" nuclear signatures, accounting for things Schwinger discussed with me.

There have been many, many important developments. Let me start with what Schwinger and Teller told me (us)

in 1989.

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Visits to me in 1989

I. Teller:

This is a near-photographic memory. It happened in the office of my boss in 1800 G street. (I actually have a video of the talk he gave at my retirement ceremony in 2015.) Teller grinned at us and said something like:

"I can tell you were smart enough to understand what was going on and what people were saying. You probably came out with the impression that there were three types of cold fusion demonstrated: (1) the Jones type, too small to be of much interest to you (DOE could handle whatever is needed for purely scientific interest); (2) the Pons and Fleischmann type, which we all know is too dangerous to get out;

and (3) the Appleby type, which seems both useful and safe. You probably want to start funding type (3).

HOWEVER... it has been my job to stay on top of the national security aspects of anything nuclear. I have been taught how to think like an enemy, terrorist or larger. To USE the Appleby technology, and halfway competent adversary will notice immediately how you can use the

Appleby type for nuclear transmutation. In other words, a huge new set of groups could breed weapons grade nuclear material on a scale which would dramatically change the world... and put the whole world at risk. PLEASE draw the obvious conclusions..."

I even got invited to one of the DIA annual big conferences, because of some discussions I had with DIA on this issue.

I have followed many of the annual LENR conferences (in part because DEnnis Bushnell, chief scientist of NASA Langley sent many people many updates). Nothing I saw there worried me... UNTIL the project at Mitsubishi crossed my desk. This is just the tip of the iceberg,

but I see no reason to say more here and now.

II. Schwinger and Pons

It is hard for me NOT to say more, because there is so much. Schwinger knew a lot more than Teller or people

like Feynmann. And I knew Schwinger much better because I had taken courses from him at Harvard, and had tea regularly with the leader of his entourage. (Among the names are https://www.nhn.ou.edu/~milton/books.html and Chaikin lead author of Principles of Condensed Matter Physics.)

Schwinger was much quieter and more scholarly than Feynmann. As a true scientist, he naturally thought about the Pons results,

and the relation to advanced topics in nuclear forces he (unlike Feynmann) had studied in very great depth. His article in Science

"A Magnetic Model of Matter" was only just an entry point to his work (and mine) on alternatives to the more popular QCD "quark model."

His models predicted long-range terms in the nuclear force which, unlike QCD, might explain the Pons results... if the results were true.

That is why he and Pons ran a very intense combined theoretical/experimental effort to find out. "We found out beyond any shadow of a doubt that it DOES work... in an apparatus which a good high school lab could use... and we all know what THAT implies."

Even before this... I had read Schwinger's paper in Science, and wondered: "WHAT does the empirical data say about QCD versus

presence of a (usable) long-range term?" I did a very thorough literature search, and was astounded to learn that the literature supported Schwinger's view over QCD, TO THE EXTENT we could tell. I contacted Tetsuo Sawada in Japan from DOE or NSF (I worked in both places in the 1980s, and have saved the documents). Their best nuclear information strengthened the story, and we wrote a joint paper which I sent to arxiv, run by LANL.

Later I learned that LANL was the place Teller probably mentioned, when he suggested USA needs SOME highly secret

R&D to stay on top of all this, just to be ready in case of need. (That reminds me of people at NSA who told me that was a major part of

their thinking in setting up an agency, then interagency program, for which I represented NSF on the coordinating group QISCOG.)

We pulled back the joint paper, though of course I still have copies of it in more than one place and so do I.

The Sawada family has remained active in many related activities, and had links to the Mitsubishi efforts, in my files and even

photos (but not the insides of any of the buildings doing nuclear things in Japan).

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I learned a whole lot after that... in more advanced and more dangerous areas. For example, when Putin proudly announced a few years ago

that Russia had a "super new nuclear technology", I understood what Nicholas Manton buried in his web page and his textbook on Topological Solitons, referring to work at Protvino which fortunately never worked. But Schwinger knew how to MAKE it work, and I visited the place in the US which has proven out prototypes of devices which can be used.

IN MY VIEW, the final section of
https://drpauljohn.blogspot.com/2025/04/from-golden-dome-to-new-jerusalem.html

points to part of why we are now called to do the RD&D to build new sensor networks capable of tracking nuclear signatures on the earth or above, even though this requires new empirical work to narrow down the choice of Lagrangians capable of fitting what Schwinger observed.